

Sharp Annular Rates Realized by Genuine Finite Spectra

Bin Wang

July 2026

Abstract

RH-306 proved that the annular mass-rate barrier is sharp for an abstract coefficient envelope but deliberately supplied no spectral realization. We close that scoped gap. The exact integer spectra of RH-316 attain the same power and logarithmic rate in both H^∞ and H^2 on every strict certified annulus. This upgrades information-class sharpness to the genuine finite normal spectral class, without claiming that the actual noisy spectrum has the sharp rate.

1 Rate and spectral approximants

Put $R = 7/5$. For a finite conjugate multiset $\mathcal{S} \subset \{|\mu| \leq q\}$, write

$$\tau_n(\mathcal{S}) = \sum_{\mu \in \mathcal{S}} \mu^n, \quad g_{\mathcal{S}}(z) = \sum_{n \geq 2} \frac{\tau_n(\mathcal{S}) - a_n}{n} z^n, \quad M_2(\mathcal{S}) = \sum_{\mu \in \mathcal{S}} |\mu|^2.$$

Let

$$\kappa(\rho) = \frac{\log(1/(q_*\rho))}{\log(q_*/q)}, \quad R < \rho < \rho_*.$$

For $X = H^\infty(\rho)$ or $H^2(\rho)$, define the finite-spectral extremal error

$$E_X(M; \rho) = \inf_{M_2(\mathcal{S}) \leq M} \|g_{\mathcal{S}}\|_X.$$

Theorem 1.1 (genuine spectral saturation). *For every fixed $R < \rho < \rho_*$ and either choice of X ,*

$$E_X(M; \rho) = \Theta_\rho \left(\frac{M^{-\kappa(\rho)}}{\log M} \right) \quad (M \rightarrow \infty).$$

The upper bound is attained, up to constants, by the exact-prefix spectra of RH-316.

Proof. RH-305 applies to every modulus-capped spectrum. If its actual mass is at most M , monotonicity of $M^{-\kappa(\rho)}/\log M$ gives the displayed lower bound for the infimum.

For the upper bound, take the RH-316 spectrum matching through order N . RH-317 gives $M_N \leq Cs^N$, where $s = q_*/q$. The target tail after the matched prefix is $O((q_*\rho)^N/N)$. For the spectral tail, the mass envelope gives

$$\sum_{n > N} \frac{|\tau_n| \rho^n}{n} \leq M_N q^{-2} \sum_{n > N} \frac{(q\rho)^n}{n} = O_\rho((q_*\rho)^N/N).$$

The same coefficient estimate gives the H^2 upper bound. Since N may be chosen as the largest integer with $Cs^N \leq M$, one has $N = (\log M)/\log s + O(1)$, and this is the displayed mass rate. $\square$

At $\rho = 1.41$, $\kappa(\rho) = 0.035045705260961\dots$

Remark 1.2. The theorem constructs genuine power sums but not the actual noisy eigenvalues. It proves spectral-class sharpness only; actual convergence and its rate remain open. Gates A–E remain false/open.